Weitong Zhang

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Professional Experience

University of North Carolina, Chapel Hill — Chapel Hill, NC, USA
Tenure-Track Assistant Professor, School of Data Science and Society — 2024–Current

Education

University of California, Los Angeles — Los Angeles, CA, USA
Ph.D. in Computer Science, Advisor: Quanquan Gu, Dissertation Year Award — 2019–2024

Tsinghua University — Beijing, China
B.E. in Automation, Outstanding Graduate Award — 2015–2019

Research Interest

My research focuses on machine learning, with an emphasis on **reinforcement learning** (RL). I aim to advance the theoretical foundations of RL and broaden its applicability including large language models, agentic systems, **embodied AI** and generative modeling. I am also working on developing **AI for scientific discovery**, particularly in the **self-driving laboratories** in chemistry using specific-designed RL algorithms and modern machine learning models.

Scholarships and Awards

- UNC Junior Faculty Award ($10,000) — 2026
- AAAI New Faculty Award — 2025
- UCLA Dissertation Year Fellowship — 2023
- UCLA Summer Mentored Research Fellowship — 2021
- Qualcomm Scholarship — 2017–2018
- Finalist in Mathematical Contests in Modeling (MCM) — 2017
- China National Scholarship — 2016
- Tsinghua Outstanding Freshmen Scholarship — 2015
- Golden Prize for Chinese Physics Olympiad (CPHO) — 2014

Research Grants and External Support

- NVIDIA Academic Program 2026, Federated Agentic Alignment: Hybrid Transfer RL with Local Self-Play. (Single PI)
- UNC – Tübingen Joint Grant: Flow-Based Generative Models for Data-Efficient Imitation Learning (Lead PI, share $10,000)
- NVIDIA Academic Program 2-25, Scalable Energy Guided Graph Diffusion Model for Accelerated Drug Design (Lead PI)
- Google Cloud Research Credit, Exact Energy Guided Generative Modeling for De-novo Drug Design

Teaching

- **Instructor** at University of North Carolina at Chapel Hill — Fall 2026
  DATA 703: Introduction to Programming (Course Website)
- **Instructor** at University of North Carolina at Chapel Hill — Spring 2026
  DATA 522: Practical Deep Learning Systems (Course Website)
- **Instructor** at University of North Carolina at Chapel Hill — Spring 2025
  DATA 890: Special Topics: Decision Making and Reinforcement Learning (Course Website)
- **Teaching Assistant** at University of California, Los Angeles — Spring 2021, Fall 2021
  Introductory Digital Design Laboratory (CS M152A) (Course Website 21 Fall, Course Website 21 Spring)
- **Teaching Assistant** at University of California, Los Angeles — Winter 2022, Winter 2023
  Fundamentals of Artificial Intelligence (CS 161) (Course Website 22 Winter, Course Website 23 Winter)

Professional Services

Conference Area Chair/Senior Program Committee

- International Conference on Machine Learning (ICML), 2025, 2026
- International Conference on Learning Representations (ICLR), 2026
- Neural Information Processing Systems (NeurIPS), 2026
- Transactions on Machine Learning Research (TMLR)

Conference Reviewer / Program Committee

- International Conference on Learning Representations (ICLR), 2020, 2021, 2022, 2023, 2024, 2025
- Neural Information Processing Systems (NeurIPS), 2021, 2022, 2023, 2024
- International Conference on Machine Learning (ICML), 2021, 2022, 2023, 2024
- International Joint Conference on Artificial Intelligence (IJCAI), 2020, 2021, 2022, 2023, 2024
- International Conference on Artificial Intelligence and Statistics (AISTATS), 2022, 2023, 2024

Journal Reviewer

- Journal of Artificial Intelligence Research (JAIR)
- Transactions on Machine Learning Research (TMLR)
- PLOS ONE
- PLOS Global Public Health

Invited Talks & Panel Discussions

- Exact Energy Guidance in Diffusion Models and offline RL
  - Invited Seminar, BCBB Seminar Series, NIH/NIEHS — 2025-11-04
  - INFORMS 2025, Emerging Technology in Integrated Risk Management — 2025-10-28
  - Research Seminar, UNC School of Dentistry, Host: Dr. Di Wu — 2025-04-13
  - Research in Progress, UNC School of Medicine — 2025-03-15
  - Samsung AI seminar — 2025-02-28

– Department of Computer Science, Indiana University Bloomington 2024-12-05
– Research Seminar, NVIDIA-Astrazeneca 2024-11-08

- Duke Industry Statistics Symposium 2025-04-10
  Panelist: S3D – Challenges and Opportunities of ML/AL in Drug Development
- Introduction to Reinforcement Learning: from Bayesian Optimization to Bandits 2024-08-29
  Research Seminar, UCLA Department of Chemistry, Host: Dr. Chong Liu
- Uncertainty-Aware Unsupervised and Robust Reinforcement Learning 2024-08-23
  Research Seminar, Department of Biostatistics & Bioinformatics, Duke University, Host: Dr. Pan Xu

Publications

* indicates equal contribution

[Hua+26] Aijian Huang, Jared S. Stanley, Yi-An Lai, Hongyuan Sheng, Haiyuan Zou, Hao Ming Chen, Cyrille Costentin, Long Qi, Jenny Y. Yang, **Weitong Zhang**, and Chong Liu. "Latent mechanism mapping in self-driving molecular electrocatalysis". In: *Submitted preprint* (2026).

[Lai+26] Yi-An Lai, Muxin Xiong, Jingwen Sun, Aijian Huang, Max Van Fleet, Hongyuan Sheng, Jennifer Kim, F. Kuo, Tai Ying Lai, Xuian-Rou Lin, Yu-An Chen, **Weitong Zhang**, Hao Ming Chen, and Chong Liu. "Chemistry-infused bandit framework for interpretable autonomous campaign of electrocatalysis". In: *Submitted preprint* (2026).

[Li+26] Shangzhe Li, Xuchao Zhang, Chetan Bansal, and **Weitong Zhang**. "Your Self-Play Algorithm is Secretly an Adversarial Imitator: Understanding LLM Self-Play through the Lens of Imitation Learning". In: *arXiv preprint arXiv:2602.01357* (2026). arXiv: 2602.01357 `[cs.LG]`.

[LZZ26] Shangzhe Li, Dongruo Zhou, and **Weitong Zhang**. "Near-Optimal Second-Order Guarantees for Model-Based Adversarial Imitation Learning". In: *International Conference on Learning Representations*. 2026.

[LW26] Shangzhe Liang and **Weitong Zhang**. "Provably Efficient Offline-to-Online Value Adaptation with General Function Approximation". In: *arXiv preprint arXiv:2604.13966* (2026). arXiv: 2604.13966 `[cs.LG]`.

[LLW26] Yiyan Liang, Sifei Liu, and **Weitong Zhang**. "Moment Matching Q-Learning". In: *International Conference on Machine Learning (ICML 2026)*. Accepted by ICML 2026. 2026. arXiv: 2605.29033 `[cs.LG]`.

[QLZ26] Yongtao Qu, Shangzhe Li, and **Weitong Zhang**. "Imitation from Observations with Trajectory-Level Generative Embeddings". In: *arXiv preprint arXiv:2601.00452* (2026).

[Sun+26] Jingwen Sun, **Weitong Zhang**, Qingyue Zhao, Quanquan Gu, and Chong Liu. "EIS-GPT: Transformer-based generation of equivalent circuit for electrochemical impedance spectroscopy". In: *ChemRxiv preprint* (2026).

[Wei26] **Weitong Zhang**. "Reinforcement Learning Without Explicit Rewards: Theory and Practice". In: *Proceedings of the AAAI Conference on Artificial Intelligence* 40.47 (2026), pp. 39847–39847.

[YWW26] Yuxiao Yang, Xiaoyun Wang, and **Weitong Zhang**. "OGLS-SD: On-Policy Self-Distillation with Outcome-Guided Logit Steering for LLM Reasoning". In: *arXiv preprint arXiv:2605.12400* (2026). arXiv: 2605.12400 [cs.LG].

[YW26] Yuxiao Yang and **Weitong Zhang**. "Return-to-Go Is More Than a Number: Q-Guided Alignment for Return-Conditioned Supervised Learning". In: *International Conference on Machine Learning (ICML 2026)*. Accepted by ICML 2026. 2026. arXiv: 2605.29028 [cs.LG].

[Yu+26a] Tianrun Yu, Yuxiao Yang, Zhaoyang Wang, Kaixiang Zhao, Porter Jenkins, Xuchao Zhang, Chetan Bansal, Huaxiu Yao, and **Weitong Zhang**. "Provable and Practical In-Context Policy Optimization for Self-Improvement". In: *International Conference on Learning Representations (ICLR 2026)*. Accepted by ICLR 2026. 2026. arXiv: 2603.01335 [cs.LG].

[Yu+26b] Tianrun Yu, Kaixiang Zhao, Chih-Chun Chen, Amanda Hughes, Taylor W Killian, Fenglong Ma, **Weitong Zhang**, and Porter Jenkins. "LARK: Learnability-Grounded Trajectory Selection for Efficient Reasoning Distillation". In: *arXiv preprint arXiv:2605.30651* (2026). arXiv: 2605.30651 [cs.LG].

[Lan+25] Yifan Lan, Yuanpu Cao, **Weitong Zhang**, Lu Lin, and Jinghui Chen. "Phi: Preference Hijacking in Multi-modal Large Language Models at Inference Time". In: *The 2025 Conference on Empirical Methods in Natural Language Processing* ***(oral)***. 2025.

[Sun+25] Jingwen Sun, **Weitong Zhang**, Yuanzhou Chen, Benjamin B Hoar, Hongyuan Sheng, Jenny Y Yang, Quanquan Gu, and Chong Liu. "Inquiry into the Appropriate Data Preprocessing of Electrochemical Impedance Spectroscopy for Machine Learning". In: *The Journal of Physical Chemistry C* (2025).

[Wan+25a] Yilin Wang, Shangzhe Li, Haoyi Niu, Zhiao Huang, **Weitong Zhang**, and Hao Su. "A Recipe for Efficient Sim-to-Real Transfer in Manipulation with Online Imitation-Pretrained World Models". In: *arXiv preprint arXiv:2510.02538* (2025).

[Wan+25b] Zhaoyang Wang, Weilei He, Zhiyuan Liang, Xuchao Zhang, Chetan Bansal, Ying Wei, **Weitong Zhang**, and Huaxiu Yao. "CREAM: Consistency Regularized Self-Rewarding Language Models". In: *International Conference on Learning Representations*. 2025.

[ZZG25] Shiyuan Zhang, **Weitong Zhang**, and Quanquan Gu. "Energy-Weighted Flow Matching for Offline Reinforcement Learning". In: *International Conference on Learning Representations*. 2025.

[Zha+25] Linxi Zhao, Yihe Deng, **Weitong Zhang**, and Quanquan Gu. "Mitigating Object Hallucination in Large Vision-Language Models via Image-Grounded Guidance". In: *Forty-second International Conference on Machine Learning* ***(spotlight)***. 2025.

[Zho+25] Yiyang Zhou, Zhaoyang Wang, Tianle Wang, Shangyu Xing, Peng Xia, Bo Li, Kaiyuan Zheng, Zijian Zhang, Zhaorun Chen, Wenhao Zheng, Xuchao Zhang, Chetan Bansal, **Weitong Zhang**, Ying Wei, Mohit Bansal, and Huaxiu Yao. "AnyPrefer: An Automatic Framework for Preference Data Synthesis". In: *International Conference on Learning Representations*. 2025.

[BZG24] Alexander Braverman, **Weitong Zhang**, and Quanquan Gu. "Mitigating Hallucination in Large Language Models with Explanatory Prompting". In: *Statistical Foundations of LLMs and Foundation Models (NeurIPS 2024 Workshop)*. 2024.

[Hoa+24] Benjamin B Hoar, **Weitong Zhang**, Yuanzhou Chen, Jingwen Sun, Hongyuan Sheng, Yucheng Zhang, Yisi Chen, Jenny Y Yang, Cyrille Costentin, Quanquan Gu, et al. "Redox-Detecting Deep Learning for Mechanism Discernment in Cyclic Voltammograms of Multiple Redox Events". In: *ACS electrochemistry* 1.1 (2024), pp. 52–62.

[Hua+24] Zijie Huang, Jeehyun Hwang, Junkai Zhang, Jinwoo Baik, **Weitong Zhang**, Dominik Wodarz, Yizhou Sun, Quanquan Gu, and Wei Wang. "Causal Graph ODE: Continuous Treatment Effect Modeling in Multi-agent Dynamical Systems". In: *Proceedings of the ACM on Web Conference 2024*. 2024, pp. 4607–4617.

[She+24] Hongyuan Sheng, Jingwen Sun, Oliver Rodríguez, Benjamin B Hoar, **Weitong Zhang**, Danlei Xiang, Tianhua Tang, Avijit Hazra, Daniel S Min, Abigail G Doyle, et al. "Autonomous closed-loop mechanistic investigation of molecular electrochemistry via automation". In: *Nature Communications* 15.1 (2024), p. 2781.

[Zha+24a] Junkai Zhang, **Weitong Zhang**, Dongruo Zhou, and Quanquan Gu. "Uncertainty-Aware Reward-Free Exploration with General Function Approximation". In: *Forty-first International Conference on Machine Learning*. 2024.

[Zha+24b] **Weitong Zhang**, Zhiyuan Fan, Jiafan He, and Quanquan Gu. "Achieving Constant Regret in Linear Markov Decision Processes". In: *The Thirty-eighth Annual Conference on Neural Information Processing Systems*. 2024.

[Zhe+24] Wenhao Zheng, Yixiao Chen, **Weitong Zhang**, Souvik Kundu, Yun Li, Zhengzhong Liu, Eric P Xing, Hongyi Wang, and Huaxiu Yao. "CITER: Collaborative Inference for Efficient Large Language Model Decoding with Token-Level Routing". In: *Adaptive Foundation Models: Evolving AI for Personalized and Efficient Learning*. 2024.

[Den+23] Yihe Deng, **Weitong Zhang**, Zixiang Chen, and Quanquan Gu. "Rephrase and respond: Let large language models ask better questions for themselves". In: *arXiv preprint arXiv:2311.04205* (2023).

[Ji+23] Kaixuan Ji, Qingyue Zhao, Jiafan He, **Weitong Zhang**, and Quanquan Gu. "Horizon-free Reinforcement Learning in Adversarial Linear Mixture MDPs". In: *The Twelfth International Conference on Learning Representations*. 2023.

[ZZG23] Junkai Zhang, **Weitong Zhang**, and Quanquan Gu. "Optimal horizon-free reward-free exploration for linear mixture mdps". In: *International Conference on Machine Learning*. PMLR. 2023, pp. 41902–41930.

[Zha+23a] **Weitong Zhang**, Jiafan He, Zhiyuan Fan, and Quanquan Gu. "On the interplay between misspecification and sub-optimality gap in linear contextual bandits". In: *International Conference on Machine Learning*. PMLR. 2023, pp. 41111–41132.

[Zha+23b] **Weitong Zhang**, Jiafan He, Dongruo Zhou, Q Gu, and A Zhang. "Provably efficient representation selection in low-rank Markov decision processes: from online to offline RL". In: *Uncertainty in Artificial Intelligence*. PMLR. 2023, pp. 2488–2497.

[Zha+23c] **Weitong Zhang**, Xiaoyun Wang, Weili Nie, Joe Eaton, Brad Rees, and Quanquan Gu. "MoleculeGPT: Instruction Following Large Language Models for Molecular Property Prediction". In: *NeurIPS 2023 Workshop on New Frontiers of AI for Drug Discovery and Development*. 2023.

[Zha+23d] **Weitong Zhang**, Xiaoyun Wang, Justin Smith, Joe Eaton, Brad Rees, and Quanquan Gu. "Diffmol: 3d structured molecule generation with discrete denoising diffusion probabilistic models". In: *ICML 2023 Workshop on Structured Probabilistic Inference & Generative Modeling*. 2023.

[Hoa+22] Benjamin B Hoar, **Weitong Zhang**, Shuangning Xu, Rana Deeba, Cyrille Costentin, Quanquan Gu, and Chong Liu. "Electrochemical mechanistic analysis from cyclic voltammograms based on deep learning". In: *ACS Measurement Science Au* 2.6 (2022), pp. 595–604.

[Jia+21] Yiling Jia, **Weitong Zhang**, Dongruo Zhou, Quanquan Gu, and Hongning Wang. "Learning Neural Contextual Bandits through Perturbed Rewards". In: *International Conference on Learning Representations*. 2021.

[ZZG21] **Weitong Zhang**, Dongruo Zhou, and Quanquan Gu. "Reward-free model-based reinforcement learning with linear function approximation". In: *Advances in Neural Information Processing Systems* 34 (2021), pp. 1582–1593.

[Wu+20] Yue Frank Wu, **Weitong Zhang**, Pan Xu, and Quanquan Gu. "A finite-time analysis of two time-scale actor-critic methods". In: *Advances in Neural Information Processing Systems* 33 (2020), pp. 17617–17628.

[Zha+20] **Weitong Zhang**, Dongruo Zhou, Lihong Li, and Quanquan Gu. "Neural Thompson Sampling". In: *International Conference on Learning Representations*. 2020.

[Zou+20] Difan Zou, Lingxiao Wang, Pan Xu, Jinghui Chen, **Weitong Zhang**, and Quanquan Gu. "Epidemic model guided machine learning for COVID-19 forecasts in the United States". In: *MedRxiv* (2020), pp. 2020–05.

[Liu+18] Shuai Liu, **Weitong Zhang**, Xiaojun Wu, Shuo Feng, Xin Pei, and Danya Yao. "A simulation system and speed guidance algorithms for intersection traffic control using connected vehicle technology". In: *Tsinghua Science and Technology* 24.2 (2018), pp. 160–170.

Publications as group authors

[Lop+24] Velma K Lopez, Estee Y Cramer, Robert Pagano, John M Drake, Eamon B O'Dea, Madeline Adee, Turgay Ayer, Jagpreet Chhatwal, Ozden O Dalgic, Mary A Ladd, et al. "Challenges of COVID-19 Case Forecasting in the US, 2020–2021". In: *PLoS computational biology* 20.5 (2024), e1011200.

[She+23] Katriona Shea, Rebecca K Borchering, William JM Probert, Emily Howerton, Tiffany L Bogich, Shou-Li Li, Willem G van Panhuis, Cecile Viboud, Ricardo Aguás, Artur A Belov, et al. "Multiple models for outbreak decision support in the face of uncertainty". In: *Proceedings of the National Academy of Sciences* 120.18 (2023), e2207537120.

[Cra+22] Estee Y Cramer, Evan L Ray, Velma K Lopez, Johannes Bracher, Andrea Brennen, Alvaro J Castro Rivadeneira, Aaron Gerding, Tilmann Gneiting, Katie H House, Yuxin Huang, et al. “Evaluation of individual and ensemble probabilistic forecasts of COVID-19 mortality in the United States”. In: *Proceedings of the National Academy of Sciences* 119.15 (2022), e2113561119.

[Bra+21] Johannes Bracher, Daniel Wolffram, Jannik Deuschel, Konstantin Görgen, Jakob L Ketterer, Alexander Ullrich, Sam Abbott, Maria Vittoria Barbarossa, Dimitris Bertsimas, Sangeeta Bhatia, et al. “A pre-registered short-term forecasting study of COVID-19 in Germany and Poland during the second wave”. In: *Nature communications* 12.1 (2021), p. 5173.

[Ray+20] Evan L Ray, Nutcha Wattanachit, Jarad Niemi, Abdul Hannan Kanji, Katie House, Estee Y Cramer, Johannes Bracher, Andrew Zheng, Teresa K Yamana, Xinyue Xiong, et al. “Ensemble forecasts of coronavirus disease 2019 (COVID-19) in the US”. In: *MedRXiv* (2020), pp. 2020–08.